

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

2. USER PERSONA DEFINITION

- **Target User:** Individuals seeking personalized fitness guidance without the financial burden of hiring a personal trainer.
- **Primary Goal:** To input basic physiological metrics and instantly receive a structured, week-long workout plan tailored to their specific body and goals.

3. JOURNEY STAGES & TOUCHPOINTS

Phase 1: Platform Onboarding

- **User Action:** The user accesses the FitBuddy web application via their browser.
- **Touchpoint:** The frontend homepage (`index.html`) featuring the primary user input form.
- **User Emotion:** Motivated but seeking a simplified, easy-to-understand process.
- **System Opportunity:** Provide a clean HTML form explicitly requesting Name, Age, Weight, Fitness Goal (e.g., Weight Loss), and Workout Intensity (e.g., Medium).

Phase 2: Data Submission & AI Processing

- **User Action:** The user clicks "Generate Plan" to submit their metrics.
- **Touchpoint:** FastAPI POST route (`/generate-workout`).
- **User Emotion:** Anticipation.
- **System Opportunity:** The backend must efficiently capture this data, orchestrate a prompt, and send it to the Google Gemini API while managing potential latency.

Phase 3: Result Rendering & Database Storage

- **System Action:** The AI returns the generated 7-day plan. Simultaneously, SQLAlchemy commits the user profile and the original AI-generated plan into the SQLite database.
- **Touchpoint:** SQLite Database (`database.py`) and Jinja2 Result UI.
- **User Emotion:** Highly satisfied and ready to start their fitness journey.
- **System Opportunity:** Render the final customized plan elegantly using Jinja2 templates so the user can easily read their daily tasks.